The second Kahn–Kalai conjecture

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Abstract

For every finite simple graph H with $h \geq 1$ edges and at most n vertices, the threshold for $G(n,p)$ to contain a copy of H is at most $C\, p_{\mathrm{E}}(n,H)(1+\log_2 h)$ with C universal, where $p_{\mathrm{E}}(n,H)$ is the least density at which every subgraph of H has expected copy count at least $1/2$. This is the second Kahn–Kalai conjecture.

1 The result

Graphs are finite and simple, $[n] = \{1,\dots,n\}$, and $G(n,p)$ is the random graph on $[n]$ in which each of the $\binom{n}{2}$ possible edges is present independently with probability p. For graphs G, F, let $N(G,F)$ be the number of subgraphs of G isomorphic to F (ordinary copies: extra edges among the chosen vertices are allowed). For a graph H with at least one edge and at most n vertices define

$$p_{\mathrm{c}}(n,H) = \inf\{p \in [0,1] : \mathbb{P}(N(G(n,p),H) > 0) \geq \tfrac{1}{2}\},$$
$$p_{\mathrm{E}}(n,H) = \inf\{p \in [0,1] : \mathbb{E}N(G(n,p),F) \geq \tfrac{1}{2} \text{ for every subgraph } F \subseteq H\}.$$

These are the *containment threshold* and the *graph expectation threshold*. Three elementary facts:

(a) $\mathbb{E}N(G(n,p),F) = N(K_n,F)\, p^{e(F)}$, by linearity over the copies of F in K_n.

(b) Both infima are attained: each constraint says that a polynomial in p is at least $1/2$, there are finitely many constraints, and $p = 1$ satisfies them. In particular

$$N(K_n,F)\, p_{\mathrm{E}}(n,H)^{e(F)} \geq \tfrac{1}{2} \qquad \text{for every subgraph } F \subseteq H. \tag{1.1}$$

(c) $0 < p_{\mathrm{E}}(n,H) \leq p_{\mathrm{c}}(n,H)$. The single-edge constraint is $\binom{n}{2}p \geq \frac{1}{2}$. At $p = p_{\mathrm{c}}$, every $F \subseteq H$ has $\mathbb{E}N(G,F) \geq \mathbb{P}(N(G,F) > 0) \geq \mathbb{P}(N(G,H) > 0) \geq \frac{1}{2}$.

Theorem 1.1. *For every $n \geq 2$ and every finite simple graph H with $h = e(H) \geq 1$ edges and at most n vertices,*

$$p_{\mathrm{c}}(n,H) \leq \min\{1,\ 2048\mathrm{e}^{50} p_{\mathrm{E}}(n,H)(1+\log_2 h)\}. \tag{1.2}$$

In particular, $p_{\mathrm{c}}(n,H) \leq 6144\mathrm{e}^{50} p_{\mathrm{E}}(n,H)\log_2 n$.

The constants are universal and H may vary with n. Kahn and Kalai use expected count 1 instead of $1/2$. By (a), for a subgraph with $g \geq 1$ edges this changes its constraint by a factor $2^{1/g} \leq 2$, so it changes p_{E} by a factor in $[1,2]$ and gives the same conjecture.

Context (not used in the proof). Thresholds for fixed targets are classical [5, 2]; here H may grow with n. Kahn and Kalai [8] conjectured that the expectation obstruction determines p_{c} up to a logarithmic factor. A logarithm is needed: for a perfect matching, $p_{\mathrm{E}} = \Theta(1/n)$ but $p_{\mathrm{c}} = \Theta(\log n/n)$ [6, 8]. Other spanning targets, such as cubes [14], have required different methods. The *first* Kahn–Kalai conjecture concerns an arbitrary increasing family $\mathcal{F}$ and its expectation threshold $q(\mathcal{F})$, defined through covers. Talagrand [15, Section 6] introduced the fractional version $q_f(\mathcal{F})$. Alweiss, Lovett, Wu and Zhang [1], building on [9], developed the fragment method. Frankston, Kahn, Narayanan and Park [7] proved $p_{\mathrm{c}}(\mathcal{F}) = O(q_f(\mathcal{F}) \log \ell)$, with ℓ bounding the size of minimal members. Park and Pham [13] proved this with q in place of q_f, and q, q_f agree up to a universal constant [12, Theorem 1.1]. For the family $\mathcal{F}_H$ of edge sets containing a copy of H one has $p_{\mathrm{E}}(n, H) \le q(\mathcal{F}_H)$, so these theorems do not bound p_{c} by p_{E}. Towards that: [11, Theorem 1] gives a single logarithm for a modified graph threshold, which equals $q_f(\mathcal{F}_H)$ [3, Equation (1)]. Dubroff, Kahn and Park [3, Theorems 1.2–1.5] proved $p_{\mathrm{c}} = O(p_{\mathrm{E}} \log^3 n)$, and the conjecture when $p_{\mathrm{E}} < 1/(3n)$; they treated cliques, cycles and bounded-degree trees as test graphs in [4, Theorems 1.4–1.5]. Tran [16] reduced the loss to $O(\log^2(2e(H)))$, proved the conjecture for trees and for graphs whose average degree is at least the logarithm of their maximum degree, and proposed a coupling conjecture [16, Section 4]. The proof below borrows two constructions. Section 2 uses the posterior resampling of [10]. Section 4 uses Tran's ratio-maximizing cores [16, Section 3], which have antecedents in [9, Claim 29] and [1, Lemma 2.4].

Proof at a glance

Key idea. A copy of H is assembled in stages $\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = H$ whose sizes grow by a factor above 17. The stages are chosen so that, given any copy of H_{i-1}, the edges added by a uniformly random extension to a copy of H_i form a σ-*spread* random set, with $\sigma = 128 p_{\mathrm{E}}$. All assembly histories form a tree whose arcs are labelled by the added edges. One random edge set W of density $\rho = 4\mathrm{e}^{50}\sigma$ is then used at *every* node of the tree at once. It replaces each label by a remainder outside W of at most half the size, and spread is kept up to a constant. After $\lfloor \log_2 h \rfloor + 1$ successful rounds all remainders are empty, so the union of the random sets contains a copy of H.

Steps and dependencies.

- Lemma 2.1 (one node): the remainder is small except with probability e^{-9m} (2.4), and choosing a given target distorts the law of W *outside that target's label* by at most e^{11m} (2.5).
- Lemma 3.2 (whole tree, one W): uses both parts of Lemma 2.1. Labels are disjoint along paths, so everything below an arc ignores W on that arc's label, and (2.5) applies.
- Theorem 3.1 (tree covering): iterates Lemma 3.2.
- Lemma 4.1 (graph tree): independent of Sections 2–3; uses (1.1).
- Theorem 1.1 is Theorem 3.1 applied to the tree of Lemma 4.1.

Conventions. log is the natural logarithm and e its base. For a finite set X and $0 \le \rho \le 1$, μ_ρ is the law of the random subset of X containing each element independently with probability ρ, and $\mu_\rho(y) = \rho^{|y|}(1-\rho)^{|X|-|y|}$ is the mass of a set y. The source reuses some letters, so we rename them. The intersection size in Lemma 2.1 is K_b with values u. An outgoing level is always j, and the summation indices in Lemma 3.2 are i, g. The size $|S|$ in Lemma 4.1 is g.

2 Resampling one spread family

Fix a finite set X. A *weighted family* consists of subsets $A_b \subseteq X$, $b \in B$, with B finite, and weights $\nu_b > 0$ with $\sum_b \nu_b = 1$. The A_b are called *labels*; repeated and empty labels are allowed. For $a > 0$ the family is *a-spread* if, for c drawn with law ν,

$$\mathbb{P}(J \subseteq A_c) \le a^{|J|} \qquad \text{for every nonempty } J \subseteq X. \tag{2.1}$$

The transition. Fix $0 < \rho < 1$ and $W \subseteq X$.

1. Choose a *source* c with law ν, and set $Y = W \cup A_c$.
2. Choose a *target* b with conditional probabilities
$$\frac{\nu_b \rho^{-|A_b|} \mathbf{1}\{A_b \subseteq Y\}}{Z(Y)}, \qquad Z(Y) = \sum_{d \in B} \nu_d \rho^{-|A_d|} \mathbf{1}\{A_d \subseteq Y\}. \tag{2.2}$$
3. Retain the *fragment* $T = A_b \setminus W$.

This is the planted-set and posterior-resampling construction of [10, Sections 2.1–2.3]. We need four facts.

(F1) *Well defined.* $Z(Y) \ge \nu_c \rho^{-|A_c|} > 0$, since $A_c \subseteq Y$.

(F2) *Inclusions.* In every transition,
$$T \subseteq A_c \cap A_b, \qquad A_b \subseteq W \cup T. \tag{2.3}$$
Indeed the target satisfies $A_b \subseteq W \cup A_c$, so $A_b \setminus W \subseteq A_c$; the rest is the definition of T.

(F3) *Locality.* Given W, the joint law of the source, the target, $Z(Y)$ and T depends on W only through $W \cap \bigcup_{b \in B} A_b$. This holds because $\mathbf{1}\{A_d \subseteq W \cup A_c\} = \mathbf{1}\{A_d \setminus A_c \subseteq W\}$ and $T = A_b \setminus W$ involve only elements of $\bigcup_b A_b$.

(F4) *Density.* If $W \sim \mu_\rho$ independently of the source, then $\mathbb{P}(Y = y) = \mu_\rho(y) Z(y)$ for every $y \subseteq X$. For a fixed source c, $Y = y$ requires $A_c \subseteq y$ and $W \setminus A_c = y \setminus A_c$, so
$$\mathbb{P}(Y = y \mid c) = \mathbf{1}\{A_c \subseteq y\}\, \rho^{|y|-|A_c|}(1-\rho)^{|X|-|y|} = \mu_\rho(y) \rho^{-|A_c|} \mathbf{1}\{A_c \subseteq y\}.$$
Averaging over $c \sim \nu$ gives $\mu_\rho(y) Z(y)$.

By (F4) and Bayes' formula, (2.2) is the posterior law of the source given Y; this explains the definition but is not used. At a *fixed* W the source has law ν exactly, while the target may not.

Lemma 2.1 (Local resampling)**.** *Suppose that $|A_b| \le m$ for all b, where m is a positive integer, that ν is a-spread, and that*
$$0 < \rho < 1, \qquad a/\rho \le \mathrm{e}^{-50}.$$
Let $W \sim \mu_\rho$, independently of the source, and perform the transition. Call it a local failure *if $Z(Y) < \mathrm{e}^{-10m}$ or $|T| > m/2$. Then*
$$\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-9m}. \tag{2.4}$$
Moreover, for each fixed $b \in B$ and every nonnegative function f depending only on $W \cap (X \setminus A_b)$,
$$\mathbb{E}\big[\mathbf{1}\{\text{target} = b,\ Z(Y) \ge \mathrm{e}^{-10m}\}\, f(W)\big] \le \mathrm{e}^{11m} \nu_b\, \mathbb{E} f(W), \tag{2.5}$$
where the right-hand expectation is over $W \sim \mu_\rho$.

Where used: in Lemma 3.2, at every node of a tree. There (2.4) bounds the failure mass, and (2.5) is applied with f the potential of the subtree below b.

Proof. **Step 0 (joint law).** By the definition of the transition, the probability that the random set, the source and the target equal a given triple (W, c, b) is

$$\mu_\rho(W)\,\nu_c\,\nu_b\,\rho^{-|A_b|}\,\frac{\mathbf{1}\{A_b \setminus A_c \subseteq W\}}{Z(W \cup A_c)}. \tag{2.6}$$

We use two remarks for fixed (c, b). First, $\mathbb{P}_W(A_b \setminus A_c \subseteq W) = \rho^{|A_b \setminus A_c|}$. Second, $\rho^{-|A_b|}\rho^{|A_b \setminus A_c|} = \rho^{-|A_c \cap A_b|}$.

Step 1 (small denominator). By (F4),

$$\mathbb{P}\big(Z(Y) < \mathrm{e}^{-10m}\big) = \sum_y \mu_\rho(y) Z(y) \mathbf{1}\{Z(y) < \mathrm{e}^{-10m}\} \le \mathrm{e}^{-10m}.$$

Step 2 (intersections). Fix b and let $K_b = |A_c \cap A_b|$ with $c \sim \nu$. For $0 \le u \le m$,

$$\mathbb{P}(K_b = u) \le \sum_{J \subseteq A_b,\ |J| = u} \mathbb{P}(J \subseteq A_c) \le \binom{m}{u} a^u. \tag{2.7}$$

The first inequality is a union bound over the possible intersections. The second is (2.1) together with $|A_b| \le m$; for $u = 0$ the bound is 1.

Step 3 (large fragment). In (2.6), bound $1/Z$ by e^{10m} on $\{Z(Y) \ge \mathrm{e}^{-10m}\}$. By (2.3), $|T| > m/2$ forces $|A_c \cap A_b| > m/2$. Summing over W with the two remarks of Step 0, and then over c using Step 2,

$$\begin{aligned}
\mathbb{P}\big(Z(Y) \ge \mathrm{e}^{-10m},\ |T| > m/2\big) &\le \mathrm{e}^{10m} \sum_{c,b} \nu_c \nu_b\, \rho^{-|A_c \cap A_b|} \mathbf{1}\{|A_c \cap A_b| > m/2\} \\
&= \mathrm{e}^{10m} \sum_b \nu_b \sum_{u > m/2} \rho^{-u}\, \mathbb{P}(K_b = u) \\
&\le \mathrm{e}^{10m} \sum_{u > m/2} \binom{m}{u} (a/\rho)^u \le \mathrm{e}^{10m} 2^m \mathrm{e}^{-25m}.
\end{aligned}$$

The last step uses $(a/\rho)^u \le \mathrm{e}^{-50u} \le \mathrm{e}^{-25m}$ for $u > m/2$ and $\sum_u \binom{m}{u} = 2^m$.

Step 4 (failure bound). By Steps 1 and 3, $\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-10m} + \mathrm{e}^{-(15 - \log 2)m}$. This is at most e^{-9m} because $\mathrm{e}^{-m} + \mathrm{e}^{-(6 - \log 2)m} \le \mathrm{e}^{-1} + \mathrm{e}^{-5} < 1$ for $m \ge 1$.

Step 5 (distortion). Fix b. Multiply (2.6) by $f(W)$, bound $1/Z$ by e^{10m}, and sum over W and c:

$$\mathbb{E}\big[\mathbf{1}\{\text{target} = b,\ Z(Y) \ge \mathrm{e}^{-10m}\} f(W)\big] \le \mathrm{e}^{10m} \nu_b \sum_c \nu_c \rho^{-|A_b|} \mathbb{E}\big[\mathbf{1}\{A_b \setminus A_c \subseteq W\} f(W)\big].$$

The indicator depends only on $W \cap A_b$ and f only on $W \setminus A_b$. Under μ_ρ these are independent, so the expectation factors as $\rho^{|A_b \setminus A_c|}\, \mathbb{E} f(W)$. By the second remark of Step 0 and then (2.7), the right-hand side equals

$$\mathrm{e}^{10m} \nu_b\, \mathbb{E} f(W) \sum_c \nu_c \rho^{-|A_c \cap A_b|} \le \mathrm{e}^{10m} \nu_b\, \mathbb{E} f(W) \sum_{u=0}^{m} \binom{m}{u} (a/\rho)^u \le \mathrm{e}^{11m} \nu_b\, \mathbb{E} f(W),$$

since $\sum_u \binom{m}{u} (a/\rho)^u = (1 + a/\rho)^m \le \mathrm{e}^m$. □

3 Covering a tree

A *probability tree on X* is a finite rooted tree whose leaves all have the same depth k, the number of *levels*. Each non-leaf node v carries a probability distribution ν_v with positive weights on its children $\mathrm{Ch}(v)$, and the arc from v to each child b carries a label $A_v(b) \subseteq X$. Level j consists of the arcs from depth $j-1$ to depth j, and a node at depth $j-1$ has *outgoing level* j. Nodes are never identified because labels agree. A *path* is a root-to-leaf path P, and $\mathcal{U}(P)$ is the union of its labels. If $k = 0$ the tree is a single node and $\mathcal{U} = \varnothing$.

The tree is *disjoint* if the labels along each path are pairwise disjoint. Level j has *capacity* m if all its labels have size at most m. It is *a-spread* if, at every node v of outgoing level j, the weighted family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v is a-spread. No spread assumption is made on the law of a whole path.

Theorem 3.1 (Tree covering)**.** *Let $\mathcal{T}$ be a disjoint probability tree on a finite set X, with $k \geq 1$ levels. Suppose that every level is σ-spread, where $\sigma > 0$, and that level i has a positive integer capacity ℓ_i with*

$$\ell_{i+1} \geq 16\ell_i \qquad (1 \leq i < k).$$

Set $r = \min\{1,\ 16\mathrm{e}^{50}\sigma(1+\log_2 \ell_k)\}$. Then

$$\mathbb{P}_{W\sim\mu_r}(\mathcal{U}(P) \subseteq W \text{ for some path } P \text{ of } \mathcal{T}) \geq \tfrac{2}{3}.$$

Where used: with the tree of Lemma 4.1, to prove Theorem 1.1.

3.1 One simultaneous reduction

Lemma 3.2 (Reduction)**.** *Let $\mathcal{T}$ be a disjoint probability tree with $k \geq 1$ levels and positive integer capacities $m_1, \dots, m_k$ satisfying $m_{j+1} \geq 16m_j$. Suppose level j is α_j-spread, where*

$$0 < \alpha_j \leq a, \qquad \rho = \mathrm{e}^{50}a < 1.$$

There is a deterministic procedure which, given $W \sim \mu_\rho$, succeeds with probability at least $3/4$ and on success returns a disjoint probability tree $\mathcal{T}'$ with the same levels, such that:

(i) level j has capacity $\lfloor m_j/2 \rfloor$ and is $\alpha_j/(1-\mathrm{e}^{-m_j})$-spread;

(ii) for every path Q of $\mathcal{T}'$ there is a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q). \tag{3.1}$$

Where used: iterated in the proof of Theorem 3.1. Output capacities may be 0 and output labels may be empty.

Proof. Plan. Run the transition of Section 2 at every node of the input tree, with the same W. A bottom-up rule declares nodes good or bad, and success means that the root is good. A potential D_v dominates badness and has small expectation. The key point (Step 4) is that the potential below a child b ignores W on the incoming label of b, so (2.5) applies. The output keeps the good children, with fragments as labels.

Step 1 (local masses). Let v have outgoing level j. Lemma 2.1 applies to the family $(A_v(b))_{b\in\mathrm{Ch}(v)}$, ν_v with $m = m_j$. The labels have size at most m_j. The family is α_j-spread, hence a-spread since $\alpha_j \leq a$. Finally $a/\rho = \mathrm{e}^{-50}$. For a fixed W, let $d_v(W)$ be the probability,

over source and target, of a local failure at v. Let $t_{vb}(W)$ be the probability of the event {no local failure, target $= b$}. Then

$$d_v(W) + \sum_{b \in \mathrm{Ch}(v)} t_{vb}(W) = 1, \qquad \mathbb{E}\, d_v(W) \le \mathrm{e}^{-9m_j}, \tag{3.2}$$

the bound being (2.4). These quantities always refer to the input tree, before any pruning.

Step 2 (good nodes). Fix W and put $\epsilon_j = \mathrm{e}^{-m_j}$. Leaves are good. Working upward, a node v of outgoing level j is *good* if

$$\sum_{b \in \mathrm{Ch}(v),\ b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j, \tag{3.3}$$

and *bad* otherwise. The procedure succeeds exactly when the root is good.

Step 3 (a potential dominating badness). Set $D_v(W) = 0$ at leaves and, at a node of outgoing level j,

$$D_v(W) = \epsilon_j^{-1}\Big(d_v(W) + \sum_{b \in \mathrm{Ch}(v)} t_{vb}(W) D_b(W)\Big). \tag{3.4}$$

Then $D_v \ge 0$, and every bad node has $D_v(W) > 1$. We prove the second claim by upward induction. If v is bad and its bad children have $D_b > 1$, the bracket is at least

$$d_v(W) + \sum_{b \text{ bad}} t_{vb}(W) = 1 - \sum_{b \text{ good}} t_{vb}(W) > \epsilon_j,$$

by (3.2) and the failure of (3.3).

Step 4 (locality of the potential). *Claim: for every child b of every node v, $D_b(W)$ is a function of $W \setminus A_v(b)$.* Let w be a non-leaf node in the subtree rooted at b. All leaves have depth k, so each arc (w, c) lies on a root-to-leaf path through the arc (v, b). By disjointness, $A_w(c) \cap A_v(b) = \varnothing$. By (F3), $d_w(W)$ and $t_{wc}(W)$ depend only on $W \cap \bigcup_c A_w(c)$, a set disjoint from $A_v(b)$. By (3.4), D_b is a function of these quantities. The claim holds for all nodes of the input tree, good or not, and labels on different branches may overlap.

Step 5 (one-step bound). Let v have outgoing level j and let $b \in \mathrm{Ch}(v)$. No local failure implies $Z(Y) \ge \mathrm{e}^{-10m_j}$, so $t_{vb}(W) \le \mathbb{P}(\text{target} = b,\ Z(Y) \ge \mathrm{e}^{-10m_j} \mid W)$. Multiply by $D_b(W) \ge 0$ and take expectations. Step 4 allows $f = D_b$ in (2.5), so

$$\mathbb{E}\big[t_{vb}(W) D_b(W)\big] \le \mathbb{E}\big[\mathbf{1}\{\text{target} = b,\ Z(Y) \ge \mathrm{e}^{-10m_j}\} D_b(W)\big] \le \mathrm{e}^{11m_j} \nu_v(b)\, \mathbb{E} D_b(W).$$

This is where the reuse of W is paid for. With (3.2) and $\epsilon_j^{-1} = \mathrm{e}^{m_j}$, the recurrence (3.4) gives

$$\mathbb{E} D_v(W) \le \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j} \sum_{b \in \mathrm{Ch}(v)} \nu_v(b)\, \mathbb{E} D_b(W). \tag{3.5}$$

Step 6 (success probability). For v of outgoing level j we claim that

$$\mathbb{E} D_v(W) \le B_j := \sum_{i=j}^{k} \exp\Big(12 \sum_{g=j}^{i-1} m_g - 8m_i\Big).$$

For $j = k$ the children are leaves and (3.5) gives $\mathrm{e}^{-8m_k} = B_k$. For $j < k$ the children have outgoing level $j+1$ and their weights sum to 1, so $\mathbb{E} D_v \le \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j} B_{j+1} = B_j$. For the root o, Step 3 and Markov's inequality give

$$\mathbb{P}(o \text{ is bad}) \le \mathbb{P}(D_o > 1) \le \mathbb{E} D_o \le B_1 = \sum_{i=1}^{k} \exp\Big(12 \sum_{g<i} m_g - 8m_i\Big).$$

Since $m_g \le 16^{-(i-g)} m_i$, we have $\sum_{g<i} m_g \le m_i/15$, so each exponent is at most $-(8-\frac{12}{15})m_i \le -7m_i$. The m_i are distinct positive integers, hence $B_1 \le \sum_{m\ge 1} \mathrm{e}^{-7m} = 1/(\mathrm{e}^7-1) < \frac{1}{4}$. The procedure therefore succeeds with probability at least $3/4$.

Step 7 (the output tree). Suppose the root is good. Starting from the root, at each good node v of outgoing level j put

$$\gamma_v(W) = \sum_{b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j > 0.$$

Keep every good child b with $t_{vb}(W) > 0$ and give it weight $t_{vb}(W)/\gamma_v(W)$. Relabel its incoming arc by $T_b = A_v(b) \setminus W$, and recurse into b. We check the properties.

- *Probability tree with the same levels.* The weights are positive and sum to 1. Every kept non-leaf node is good, so $\gamma_v > 0$ and it has a kept child. Hence all kept leaves are original leaves, at depth k.
- *Disjoint.* The new labels are subsets of the old ones on the same arcs.
- *Recovery* (3.1). A path Q of $\mathcal{T}'$ uses the arcs of a path P of $\mathcal{T}$, and $A_v(b) \subseteq W \cup T_b$ on each arc.
- *Capacity.* If $t_{vb}(W) > 0$, some transition at v with target b is not a local failure. Its fragment is T_b, so $|T_b| \le m_j/2$.
- *Deterministic.* $\mathcal{T}'$ is a function of W through the numbers $t_{vb}(W)$. Sources and targets serve only the analysis.

Step 8 (spread of the output). Let v be a kept node of outgoing level j and let $J \subseteq X$ be nonempty. Under the new child law, the probability that J lies in the label is

$$\frac{1}{\gamma_v(W)} \sum_{b \text{ good},\ J \subseteq T_b} t_{vb}(W).$$

The sum is the probability, given W, that the transition at v has no local failure, has a good target, and has $J \subseteq T$. By (2.3), $T \subseteq A_v(c)$ for the source c, and the source has law ν_v whatever W is. So the sum is at most $\mathbb{P}_{c\sim\nu_v}(J \subseteq A_v(c)) \le \alpha_j^{|J|}$, and

$$\frac{\alpha_j^{|J|}}{\gamma_v(W)} \le \frac{\alpha_j^{|J|}}{1-\epsilon_j} \le \Big(\frac{\alpha_j}{1-\epsilon_j}\Big)^{|J|} \qquad (|J| \ge 1).$$

This proves (i), with $\epsilon_j = \mathrm{e}^{-m_j}$. □

3.2 Iteration

Proof of Theorem 3.1. Plan. Apply Lemma 3.2 repeatedly with fresh random sets. Each success halves all capacities. After s successes all labels are empty, and the union of the sets used contains a path union of $\mathcal{T}$. Then count attempts.

Step 1 (parameters). If $r = 1$ then $W = X$ and the claim is trivial. Let $r < 1$ and

$$a = 4\sigma, \qquad \rho = \mathrm{e}^{50} a, \qquad s = \lfloor \log_2 \ell_k \rfloor + 1.$$

Then $0 < \rho = 4\mathrm{e}^{50}\sigma \le 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k) < 1$.

Step 2 (the process and its invariant). Let $W_1, W_2, \dots$ be independent with law μ_ρ. Attempt number x applies the procedure of Lemma 3.2 to the current tree with the set W_x. If the attempt fails, the tree is unchanged. If it succeeds, the tree is replaced by the output, truncated as described below. Let $\mathcal{T}_t$ be the tree after t successes, with $\mathcal{T}_0 = \mathcal{T}$, and for $0 \le t \le s$ put

$$m_i(t) = \Big\lfloor \frac{\ell_i}{2^t} \Big\rfloor, \qquad \alpha_i(t) = \sigma \prod_{u=0}^{t-1} \big(1 - \mathrm{e}^{-\lfloor \ell_i/2^u \rfloor}\big)^{-1}. \tag{3.6}$$

Invariant. $\mathcal{T}_t$ is a disjoint probability tree whose levels are the original levels i with $m_i(t) \ge 1$. Level i has capacity $m_i(t)$ and is $\alpha_i(t)$-spread. For every path Q of $\mathcal{T}_t$ there is a path P of $\mathcal{T}$ such that $\mathcal{U}(P) \subseteq \mathcal{U}(Q) \cup$ (the union of the sets used in the first t successful attempts).

The invariant holds for $t = 0$. Assume it for some $t < s$. Then $2^t \le \ell_k$, so at least one level remains. The remaining capacities satisfy $m_{i+1}(t) \ge \lfloor 16\ell_i/2^t \rfloor \ge 16 m_i(t)$, and $\alpha_i(t) \le a$ by Step 3. So Lemma 3.2 applies to $\mathcal{T}_t$. On success, its output has capacities $\lfloor m_i(t)/2 \rfloor = m_i(t+1)$, because $\lfloor \lfloor x \rfloor /2 \rfloor = \lfloor x/2 \rfloor$. Its spread parameters are $\alpha_i(t)/(1 - \mathrm{e}^{-m_i(t)}) = \alpha_i(t+1)$. Composing (3.1) with the invariant at t gives the path statement at $t+1$.

Truncation. The ℓ_i increase, so the levels with $m_i(t+1) = 0$ form an initial segment, and all their labels are empty. Follow, by a fixed rule, a path from the root through this segment. Let $\mathcal{T}_{t+1}$ be the subtree hanging from its endpoint; if no level remains, this is a single node. Every path Q' of $\mathcal{T}_{t+1}$, prefixed by the chosen arcs, is a path Q of the output with $\mathcal{U}(Q) = \mathcal{U}(Q')$. Disjointness, capacities and spread concern single paths or single nodes, so they survive. The invariant holds at $t+1$.

Step 3 (spread stays below a). For a level i present in $\mathcal{T}_t$, the integers $\lfloor \ell_i/2^u \rfloor$, $0 \le u < t$, are positive. They are distinct, since $\lfloor x/2 \rfloor < x$ for integers $x \ge 1$. Using $-\log(1-x) \le x/(1-x)$ with $x = \mathrm{e}^{-m} \le \mathrm{e}^{-1}$,

$$\sum_{m=1}^{\infty} -\log(1 - \mathrm{e}^{-m}) \le \frac{1}{1 - \mathrm{e}^{-1}} \sum_{m=1}^{\infty} \mathrm{e}^{-m} = \frac{\mathrm{e}^{-1}}{(1 - \mathrm{e}^{-1})^2} < 0.93 < \log 4.$$

Hence $\prod_{m \ge 1} (1 - \mathrm{e}^{-m})^{-1} < 4$ and $\alpha_i(t) \le 4\sigma = a$.

Step 4 (completion). Since $2^s > \ell_k \ge \ell_i$, we have $m_i(s) = 0$ for all i. So $\mathcal{T}_s$ is a single node whose only path has empty union. By the invariant, some path P of $\mathcal{T}$ has $\mathcal{U}(P)$ contained in the union of the sets used in the s successful attempts.

Step 5 (number of attempts). Given all earlier sets, the current tree is fixed and the next set has law μ_ρ. By Lemma 3.2 and Steps 2–3, the next attempt succeeds with conditional probability at least $3/4$. So the waiting time τ from one success to the next satisfies $\mathbb{P}(\tau > u) \le 4^{-u}$, and $\mathbb{E}\tau \le \sum_{u \ge 0} 4^{-u} = 4/3$. The number N of attempts needed for s successes has $\mathbb{E}N \le 4s/3$. By Markov's inequality, $\mathbb{P}(N > 4s) \le 1/3$.

Step 6 (density). Let $W_* = W_1 \cup \dots \cup W_{4s}$. Each element of X is absent from W_* independently with probability $(1-\rho)^{4s}$, so $W_* \sim \mu_{r_*}$ with

$$r_* = 1 - (1-\rho)^{4s} \le 4s\rho = 16\mathrm{e}^{50}\sigma s \le 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k) = r.$$

On $\{N \le 4s\}$, an event of probability at least $2/3$, Step 4 gives a path P with $\mathcal{U}(P) \subseteq W_*$. The event "$W$ contains some $\mathcal{U}(P)$" is increasing in W and $r_* \le r$, so the bound holds for μ_r. $\square$

4 Application to graph containment

We construct the tree needed in Theorem 3.1, following the normalized-containment maximization of Tran [16, Section 3, Lemma 3.1] with constants adapted to the factor 16.

Conventions. Put $q = p_{\mathrm{E}}(n, H)$, defined from the original H, and $\sigma = 128q$. Delete the isolated vertices of H and of every graph below, and identify a graph with its edge set. A *copy* of an edge set F is a subset of $X = \binom{[n]}{2}$ isomorphic to it. The deletion does not change the event of containing H. Copies need not be induced and $n \geq |V(H)|$, so a copy of the non-isolated part extends by adding unused vertices. For edge sets F, I let

$$M(F) = N(K_n, F), \qquad M(I, F) = N(I, F), \qquad M(\varnothing) = M(I, \varnothing) = 1.$$

These count copies, not vertex embeddings: $M(F) = (n)_{v(F)}/|\operatorname{Aut}(F)|$, a formula we do not need. Every edge subset $F \subseteq H$ is, on its incident vertices, a subgraph of the original H. So by (1.1) and fact (a),

$$M(F)\, q^{|F|} \geq \tfrac{1}{2} \qquad (F \subseteq H). \tag{4.1}$$

Copy-ratio identity (also [11, Equation (5)]). Let $\boldsymbol{I}$ be a uniformly random copy of I and let F_0 be a fixed copy of an edge subset $F \subseteq I$. Then

$$\mathbb{P}(F_0 \subseteq \boldsymbol{I}) = \frac{M(I, F)}{M(F)}. \tag{4.2}$$

Indeed, the law of $\boldsymbol{I}$ is invariant under permutations of $[n]$, so the probability is the same for all $M(F)$ copies F_0. Their sum is the expected number of copies of F inside $\boldsymbol{I}$, which is $M(I, F)$.

Lemma 4.1 (Graph hierarchy)**.** *There is a disjoint probability tree on $X = \binom{[n]}{2}$ whose path unions are copies of H with its isolated vertices removed, such that every level is σ-spread, all labels of level i have the same size ℓ_i, and*

$$1 \leq \ell_1, \qquad \ell_{i+1} \geq 16\ell_i, \qquad \ell_k \leq e(H).$$

Proof. Plan. Define a "core" of an edge set I by maximizing a normalized containment probability. Show that the core is much smaller than I (Step 1), iterate to get a chain (Step 2), turn the chain into a tree (Step 3), and read spread off the maximality (Step 4).

Step 1 (cores are small). For a nonempty edge set $I \subseteq H$ and an edge subset $S \subseteq I$, let

$$R_\sigma(I, S) = \sigma^{-|S|}\, \mathbb{P}(S_0 \subseteq \boldsymbol{I}), \tag{4.3}$$

where S_0 is any copy of S; by symmetry the choice is immaterial. Choose, by a fixed tie-breaking rule, an S maximizing $R_\sigma(I, S)$. Since $R_\sigma(I, \varnothing) = 1$, the maximum is at least 1. Now let $g = |S| > 0$ and $h_I = |I|$. The copies of S in I are g-subsets of I, so $M(I, S) \leq \binom{h_I}{g}$, and $1/M(S) \leq 2q^g$ by (4.1). By (4.2),

$$R_\sigma(I, S) \leq 2\binom{h_I}{g}(q/\sigma)^g = 2\binom{h_I}{g}128^{-g}.$$

If $g \geq h_I/17$, then $\binom{h_I}{g} \leq (\mathrm{e}h_I/g)^g \leq (17\mathrm{e})^g$, and $R_\sigma(I, S) \leq 2(17\mathrm{e}/128)^g < 2 \cdot 0.37 < 1$. So such an S is not a maximizer, and every maximizer satisfies $|S| < |I|/17$.

Step 2 (the chain). Start from $I = H$ and replace I by its maximizer until the empty set is reached. Reversing the order,

$$\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = H, \qquad |H_{i-1}| < |H_i|/17, \qquad k \geq 1,$$

where H_{i-1} maximizes $R_\sigma(H_i, \cdot)$. Put $\ell_i = |H_i| - |H_{i-1}| > 0$. Then $\ell_1 \geq 1$, $\ell_k \leq |H| = e(H)$ and $\ell_{i+1} > 17|H_i| - |H_i| = 16|H_i| \geq 16\ell_i$.

Step 3 (the tree). The root is $\varnothing$, the copy of H_0. A node at depth $i-1$ is reached through a history ending in a copy S_0 of H_{i-1}. Its children are all copies I_0 of H_i with $I_0 \supseteq S_0$, with the uniform distribution, and the arc is labelled $I_0 \setminus S_0$. Distinct histories are distinct nodes. Children exist: S_0 is the image of H_{i-1} under an injection $V(H_{i-1}) \to [n]$, which extends to an injection of $V(H_i)$ because $|V(H_i)| \leq n$. Along a path $\varnothing \subset S_1 \subset \cdots \subset S_k$ the labels $S_i \setminus S_{i-1}$ are disjoint, have sizes ℓ_i, and have union S_k, a copy of H.

Step 4 (spread). Fix a node with copy S_0 of H_{i-1}, let A be its random label, and let $J \subseteq X$ be nonempty. If $J \cap S_0 \neq \varnothing$, then $\mathbb{P}(J \subseteq A) = 0$. Otherwise, the child is a uniform copy $\boldsymbol{I}$ of H_i conditioned on $S_0 \subseteq \boldsymbol{I}$, so

$$\mathbb{P}(J \subseteq A) = \frac{\mathbb{P}(S_0 \cup J \subseteq \boldsymbol{I})}{\mathbb{P}(S_0 \subseteq \boldsymbol{I})}. \tag{4.4}$$

If the numerator is 0 we are done. If it is positive, $S_0 \cup J$ lies in some copy of H_i, so it is a copy of an edge subset $U \subseteq H_i$ with $|U| = |S_0| + |J|$. Maximality of H_{i-1} gives $R_\sigma(H_i, U) \leq R_\sigma(H_i, H_{i-1})$, that is,

$$\sigma^{-(|S_0|+|J|)}\mathbb{P}(S_0 \cup J \subseteq \boldsymbol{I}) \leq \sigma^{-|S_0|}\mathbb{P}(S_0 \subseteq \boldsymbol{I}).$$

With (4.4) this is $\mathbb{P}(J \subseteq A) \leq \sigma^{|J|}$. □

Proof of Theorem 1.1. Apply Theorem 3.1 to the tree of Lemma 4.1, with $\sigma = 128q$ and capacities ℓ_i. A random edge set of density $r = \min\{1, 16\mathrm{e}^{50} \cdot 128\, q(1 + \log_2 \ell_k)\}$ contains some path union, that is, a copy of H, with probability at least $2/3 \geq 1/2$. Since $\ell_k \leq h$ and $16 \cdot 128 = 2048$, we get $r \leq r' := \min\{1, 2048\mathrm{e}^{50} q(1 + \log_2 h)\}$. Containment is monotone in the density, so $p_{\mathrm{c}}(n, H) \leq r'$, which is (1.2). Finally $h \leq \binom{n}{2} \leq n^2$ and $\log_2 n \geq 1$ give $1 + \log_2 h \leq 1 + 2\log_2 n \leq 3\log_2 n$, and $3 \cdot 2048 = 6144$. □

Verification checklist

1. **Independence behind** (2.5) (Lemma 2.1, Steps 0 and 5). The joint law (2.6) must be right. After bounding $1/Z$, the only remaining dependence on W is the event $A_b \setminus A_c \subseteq W$, which lives on A_b, while f lives off A_b.
2. **Locality of the potential** (Lemma 3.2, Step 4). This is the one place where disjointness along paths enters the probability estimate. It licenses $f = D_b$ in Step 5, and Step 5 is what allows one W to serve all nodes.
3. **Arithmetic of the recursion** (Lemma 3.2, Step 6). The loss e^{12m_j} per level must be beaten by the gain e^{-8m_i} at deeper levels. This needs $\sum_{g<i} m_g \leq m_i/15$, which is the role of the factor 16.
4. **Spread survives conditioning** (Lemma 3.2, Step 8; Theorem 3.1, Step 3). The fragment lies in the *source* label, whose law is ν_v at fixed W. The cost is $1/(1 - \mathrm{e}^{-m_j})$ per round, and the product over rounds is below 4 because the capacities met are distinct integers.

5. **Bookkeeping of the iteration** (Theorem 3.1, Steps 2, 4–6). Check the invariant with truncation of zero-capacity levels, the count $s = \lfloor \log_2 \ell_k \rfloor + 1$, Markov with $4s$ attempts, and $r_* \le 4s\rho$.

6. **Graph input** (Lemma 4.1, Steps 1 and 4). Check that (4.1) uses the attained infimum (1.1), that $2(17\mathrm{e}/128)^g < 1$, and that maximality is applied to $U \cong S_0 \cup J$.

7. **Constants.** $a = 4\sigma$, $\rho = \mathrm{e}^{50}a$, $4s$ attempts and $\sigma = 128q$ give $4 \cdot 4 \cdot 128 = 2048$.

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